

Homework 02 Solutions

Question 1: Training and test error under a fixed design

Solution

R

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```
j  train theory_train  test theory_test first_test
1  0 1.3054      1.2974 1.331      1.317      1.370
6  5 0.9817      0.9750 1.090      1.095      1.223
11 10 0.8977      0.8992 1.120      1.119      1.181
16 15 0.8414      0.8419 1.163      1.162      1.290
21 20 0.7882      0.7900 1.218      1.210      1.363
```

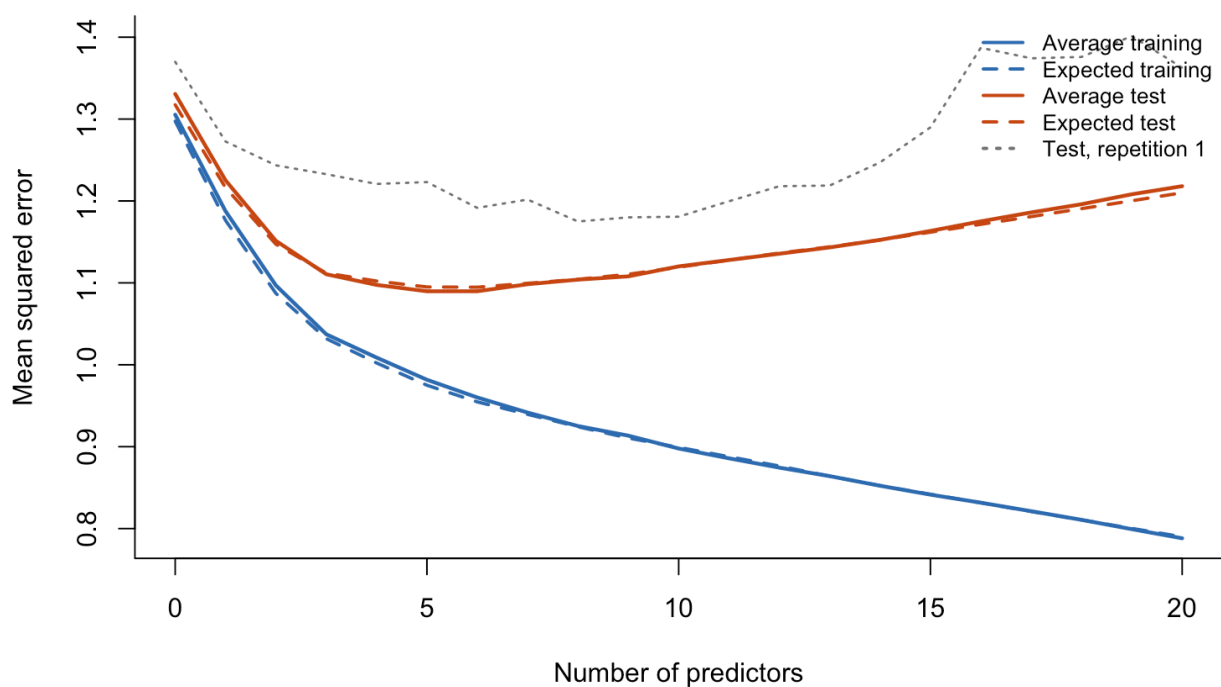
► Show the solution code

```
training_is_monotone
      TRUE
```

► Show the solution code

```
minimum_average_test minimum_expected_test
                  5                  6
```

► Show the solution code



Training and test MSE along the nested model sequence.

The simulation averages should be close to their theoretical expectations, apart from Monte Carlo error. The training curve decreases because every larger model contains the previous model.

Expected test MSE contains the mean squared approximation bias B_j^2/n , which generally decreases, and the estimation-variance term d_j/n , which increases. Their competition can produce an interior minimum. A single test-MSE curve is irregular because it retains the random variation from one response vector.

Expected training MSE contains $-d_j/n$ instead of $+d_j/n$. It therefore rewards the flexibility that allows a model to follow training noise and cannot select the predictor count honestly by itself.

Question 2: Prediction error at one target point

R

► Show the solution code

⋮

Solution

The target mean is

$$\mu_0 = \mathbf{x}_0^\top \boldsymbol{\beta} = 0.5 + 0.5 = 1.$$

Only coordinates 2 and 5 of $\mathbf{x}_0$ are nonzero. The theoretical values are:

j	Squared target-point bias	Estimation variance	Expected squared error
0	1.00	0.01	1.01
1	1.00	0.01	1.01
2	0.25	0.02	0.27
3	0.25	0.02	0.27
4	0.25	0.02	0.27
5	0.00	0.03	0.03
6	0.00	0.03	0.03

The smallest predictor count that minimizes the expected error is $j = 5$.

R

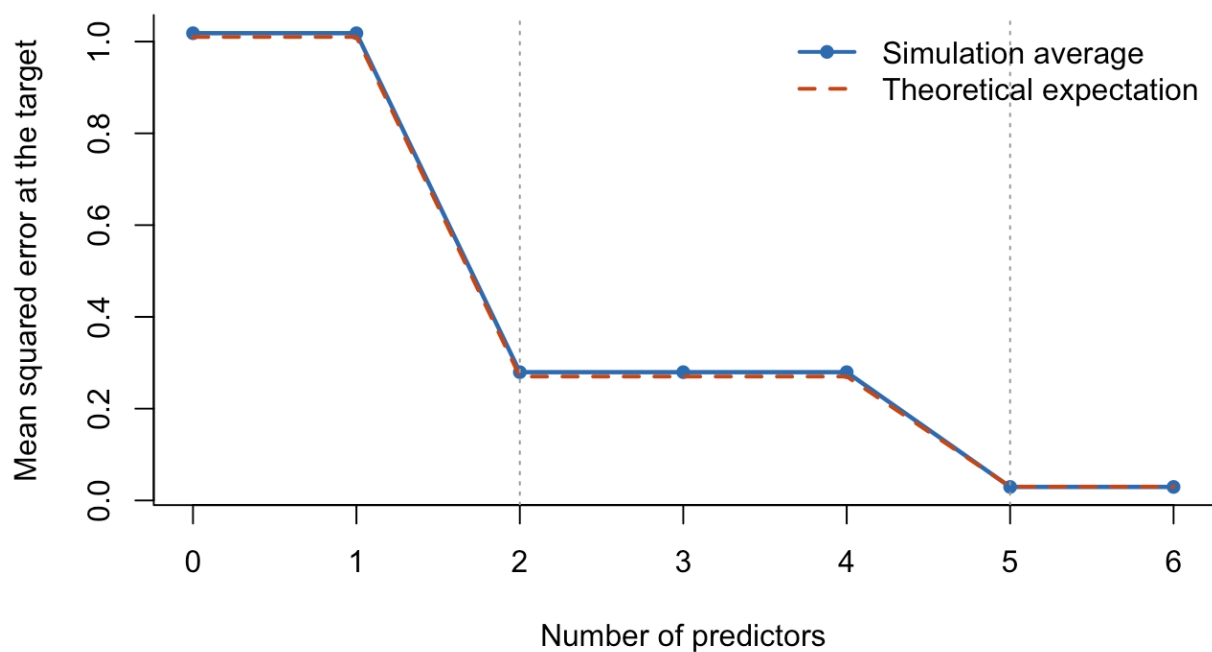
► Show the solution code

```
j simulated theoretical
1 0 1.0182 1.01
2 1 1.0182 1.01
3 2 0.2795 0.27
4 3 0.2795 0.27
5 4 0.2795 0.27
6 5 0.0294 0.03
7 6 0.0294 0.03
```

► Show the solution code

```
first_minimum
5
```

► Show the solution code



Squared error at the fixed target point.

The error changes only when predictor 2 or 5 is added. The other target coordinates are zero, so their fitted coefficients are multiplied by zero in this prediction.

This is a target-specific conclusion. Average test MSE over the entire design gives a predictor weight wherever its values are nonzero across the design rows. A predictor can therefore matter for average prediction even when it does not contribute at one particular target.

Question 3: The optimism correction

Solution

The training residual is

$$\begin{aligned}\mathbf{y} - \mathbf{H}\mathbf{y} &= (\mathbf{I}_n - \mathbf{H})(\boldsymbol{\mu} + \boldsymbol{\epsilon}) \\ &= (\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.\end{aligned}$$

The cross term has expectation zero. Since $\mathbf{I}_n - \mathbf{H}$ is symmetric and idempotent,

$$\begin{aligned}E\left(\|\mathbf{y} - \mathbf{H}\mathbf{y}\|_2^2 \mid \mathbf{Z}\right) &= B^2 + \sigma^2 \text{tr}(\mathbf{I}_n - \mathbf{H}) \\ &= B^2 + \sigma^2(n - d).\end{aligned}$$

Dividing by n gives the stated expected training MSE.

For the independent test response,

$$\mathbf{y}^* - \mathbf{H}\mathbf{y} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu} + \boldsymbol{\epsilon}^* - \mathbf{H}\boldsymbol{\epsilon}.$$

All cross terms again have expectation zero. Independence gives

$$\begin{aligned}E\left(\|\boldsymbol{\epsilon}^* - \mathbf{H}\boldsymbol{\epsilon}\|_2^2 \mid \mathbf{Z}\right) &= n\sigma^2 + \sigma^2 \text{tr}(\mathbf{H}) \\ &= \sigma^2(n + d).\end{aligned}$$

Therefore,

$$E(\text{MSE}_{\text{test}} \mid \mathbf{Z}) = \frac{B^2}{n} + \sigma^2 \left(1 + \frac{d}{n}\right).$$

Subtracting the expectations gives

$$E(\text{MSE}_{\text{test}} - \text{MSE}_{\text{train}} \mid \mathbf{Z}) = \frac{2d\sigma^2}{n}.$$

For the stated values,

$$E(\text{MSE}_{\text{train}} \mid \mathbf{Z}) = 0.04 + 1 - \frac{5}{120} = 0.998333,$$

$$E(\text{MSE}_{\text{test}} \mid \mathbf{Z}) = 0.04 + 1 + \frac{5}{120} = 1.081667,$$

and their difference is 0.083333. These are long-run averages, so a particular test MSE can still be smaller than its corresponding training MSE.

The corrected estimate is

$$\widehat{\text{MSE}}_{\text{test}} = \frac{116.4 + 2(5)(0.96)}{120} = 1.05.$$

Mallows' C_p is

$$C_p = \frac{116.4}{0.96} - 120 + 2(5) = 11.25.$$

Their relationship is

$$\begin{aligned} \frac{\hat{\sigma}^2(C_p + n)}{n} &= \frac{0.96(11.25 + 120)}{120} \\ &= 1.05 \\ &= \frac{\text{RSS} + 2d\hat{\sigma}^2}{n}. \end{aligned}$$

When n and $\hat{\sigma}^2$ are common to all candidates, this is an increasing transformation. The two criteria therefore rank the models in the same order.

Question 4: Comparing Mallows' C_p , AIC, and BIC

Solution

The full model is used only to estimate one common noise variance. Using a different variance estimate for each candidate would change the C_p comparison.

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```
residual_df.Full sigma2_full.Full
      359.000      3034.684
```

► Show the solution code

	model	d	rss	cp	aic	bic
1	Model A	8	1098909	8.116253	2974.640	3005.948
2	Model B	9	1089717	7.087434	2973.532	3008.754

► Show the solution code

aic_bic_fit_gain.A	cp_fit_gain.A	aic_penalty	bic_penalty
3.107775	3.028819	2.000000	5.913503
cp_penalty			
2.000000			

The full model has

$$n - 11 = 359$$

residual degrees of freedom. Therefore,

$$\hat{\sigma}_{\text{full}}^2 = 3034.684460.$$

The candidate results are:

Model	d	RSS	C_p	Reduced AIC	Reduced BIC
Model A	8	1,098,908.566	8.116	2,974.640	3,005.948
Model B	9	1,089,717.057	7.087	2,973.532	3,008.754

Mallows' C_p and AIC select Model B, while BIC selects Model A. The improvement in the AIC and BIC fit term is

$$n \log \left(\frac{\text{RSS}_A}{\text{RSS}_B} \right) = 3.108.$$

This exceeds AIC's additional penalty of 2, but is smaller than BIC's additional penalty $\log(n) = 5.914$. On the C_p scale,

$$\frac{\text{RSS}_A - \text{RSS}_B}{\hat{\sigma}_{\text{full}}^2} = 3.029 > 2.$$

Thus, AIC and C_p accept the improvement from adding **s6**, while BIC does not. This is a comparison of two candidates under particular assumptions. It does not prove that either candidate is the true data-generating model.

Question 5: Validation and final test data

Solution

Step 2 is the first misuse. The final test responses are used to compare the candidates, so predictor count is selected using the final test data even though the coefficients were fitted using only the training data.

Each test MSE contains random variation. Taking the minimum tends to select a model whose test MSE was unusually small by chance. Reporting the same minimum as the final assessment therefore gives an optimistic estimate of future prediction error.

A valid procedure is:

1. Reserve rows 371 through 442 without inspecting their responses.
2. Divide rows 1 through 370 into five folds.
3. For each predictor count, fit on four folds and calculate MSE on the remaining fold. Repeat until every fold has served once as the validation fold.
4. Average the five validation MSE values and select the predictor count with the smallest average.
5. Refit the selected model using all 370 training observations.
6. Evaluate this fitted model once on rows 371 through 442.

Any data-dependent preprocessing must also be estimated within the four training folds. If all eleven final test MSE values have already been examined, the analysis can still be reported as exploratory, but the reported minimum is not an untouched final assessment. A new independent test dataset or genuinely future observations are needed for a new final evaluation.

